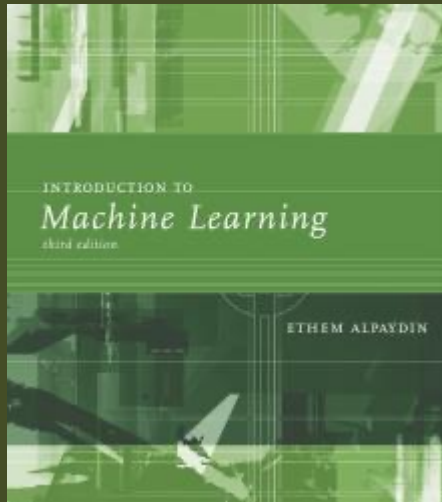




Lecture Slides for

INTRODUCTION TO MACHINE LEARNING

3RD EDITION

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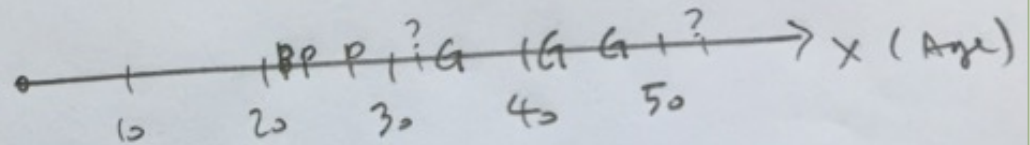
CHAPTER 3:

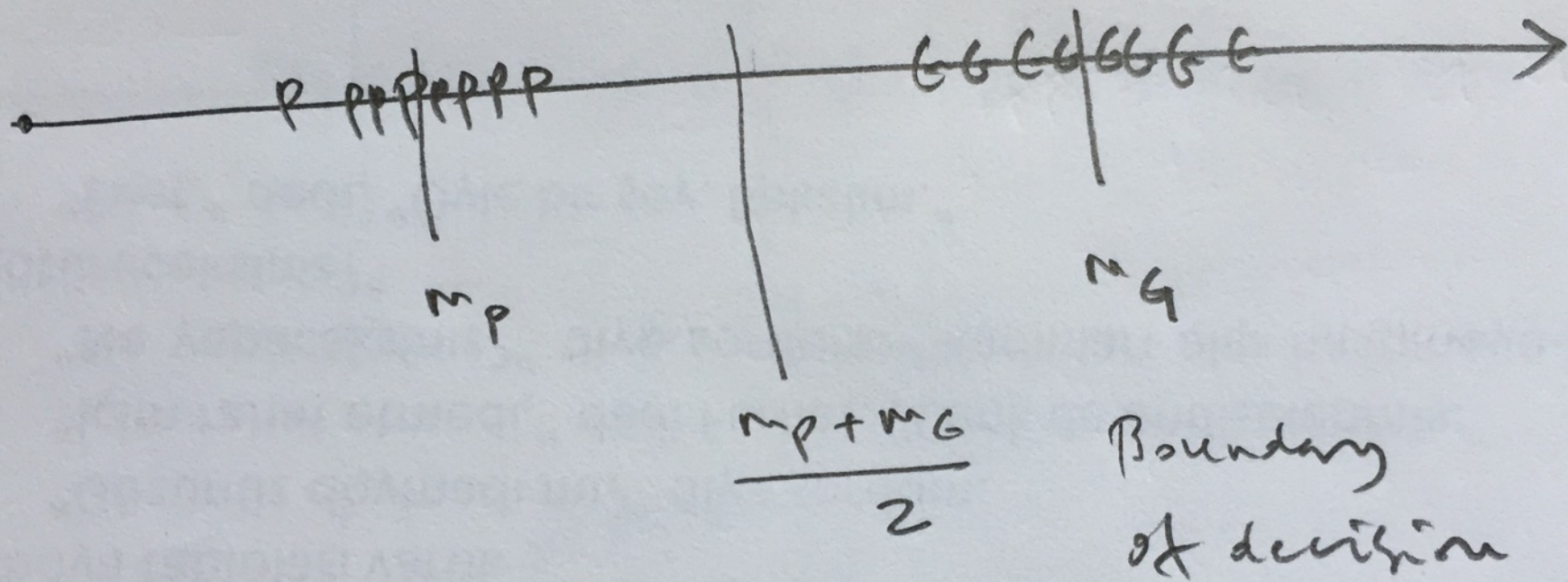
BAYESIAN DECISION THEORY

Classification

(1)

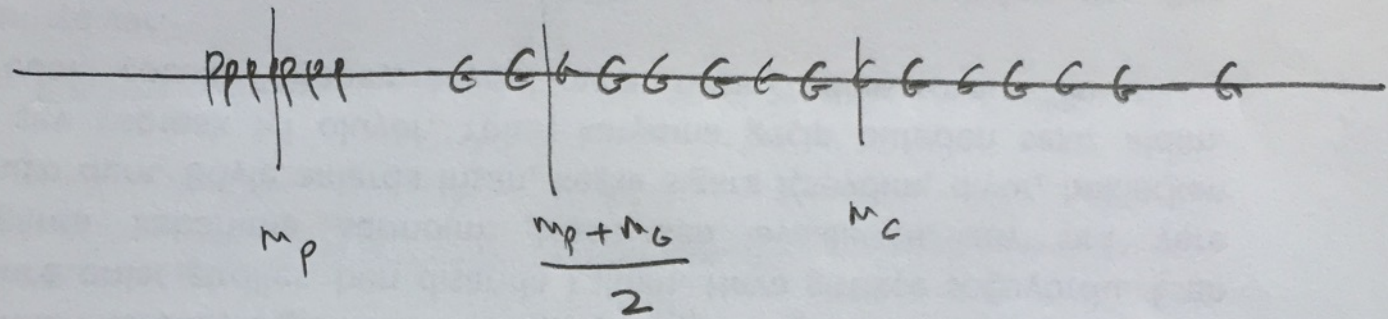
Age	Car type
21	Polo
22	Polo
35	Golf
22	Polo
41	Golf
46	Golf
33	?
52	?



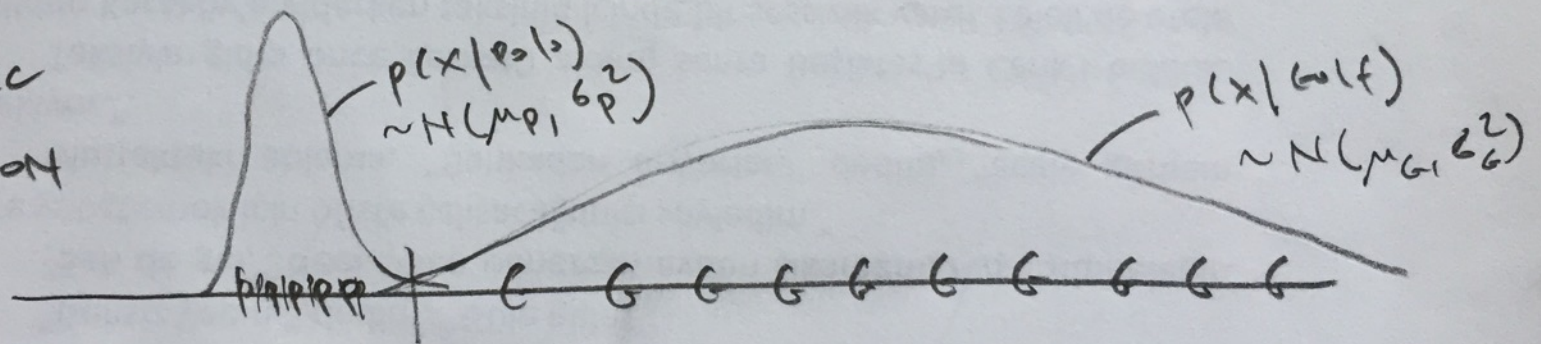


③

Doesn't work!



PROBABILISTIC
CLASSIFICATION



$$p(x|Polo) = p(x|Golf)$$

$$p(Polo|x) = \frac{p(x|Polo)p(Polo)}{p(x) = p(x|Polo)p(Polo) + p(x|Golf)p(Golf)}$$

Probability and Inference

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- Result of tossing a coin is $\in \{\text{Heads}, \text{Tails}\}$

- Random var $X \in \{1, 0\}$

Bernoulli: $P\{X=1\} = p_o$, $P\{X=0\} = 1 - p_o$

$$P\{X\} = p_o^X (1 - p_o)^{(1-X)}$$

- Sample: $\mathbf{X} = \{x^t\}_{t=1}^N$

Estimation: $p_o = \#\{\text{Heads}\} / \#\{\text{Tosses}\} = \sum_t x^t / N$

- Prediction of next toss:

Heads if $p_o > 1/2$, Tails otherwise

Classification

- Credit scoring: Inputs are income and savings.

Output is low-risk vs high-risk

- Input: $\mathbf{x} = [x_1, x_2]^T$, Output: $C \in \{0, 1\}$

- Prediction:

choose $\begin{cases} C = 1 \text{ if } P(C = 1 | x_1, x_2) > 0.5 \\ C = 0 \text{ otherwise} \end{cases}$

or

choose $\begin{cases} C = 1 \text{ if } P(C = 1 | x_1, x_2) > P(C = 0 | x_1, x_2) \\ C = 0 \text{ otherwise} \end{cases}$

Bayes' Rule

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The diagram shows the Bayes' Rule formula with four labels and arrows pointing to specific parts of the equation:

- posterior*: An arrow points to $P(C | \mathbf{x})$.
- prior*: An arrow points to $P(C)$.
- likelihood*: An arrow points to $p(\mathbf{x} | C)$.
- evidence*: An arrow points to $p(\mathbf{x})$.

$$P(C | \mathbf{x}) = \frac{P(C) p(\mathbf{x} | C)}{p(\mathbf{x})}$$

$$P(C = 0) + P(C = 1) = 1$$

$$p(\mathbf{x}) = p(\mathbf{x} | C = 1)P(C = 1) + p(\mathbf{x} | C = 0)P(C = 0)$$

$$p(C = 0 | \mathbf{x}) + p(C = 1 | \mathbf{x}) = 1$$

Bayes' Rule: $K > 2$ Classes

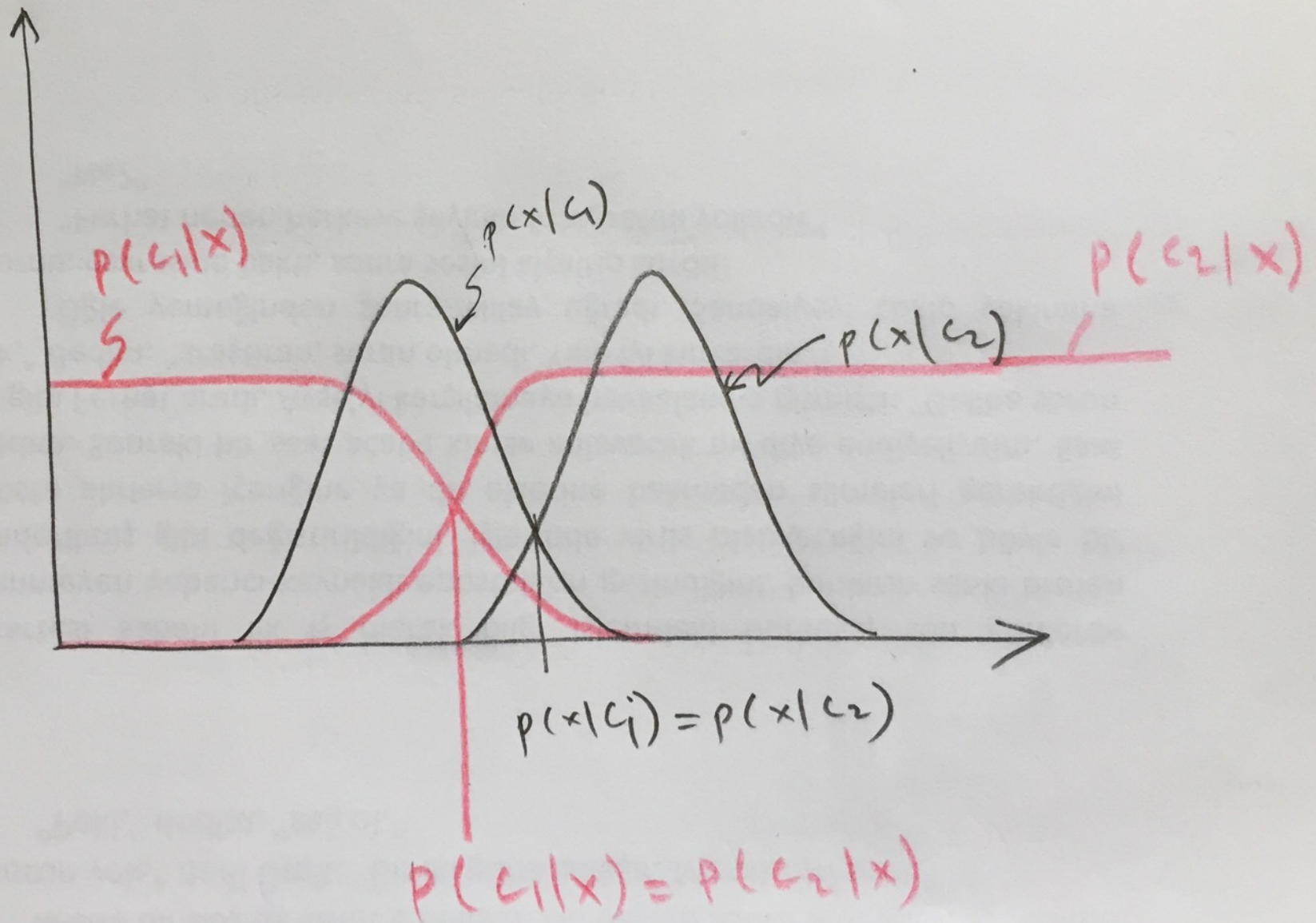
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$$\begin{aligned} P(C_i | \mathbf{x}) &= \frac{p(\mathbf{x} | C_i)P(C_i)}{p(\mathbf{x})} \\ &= \frac{p(\mathbf{x} | C_i)P(C_i)}{\sum_{k=1}^K p(\mathbf{x} | C_k)P(C_k)} \end{aligned}$$

$$P(C_i) \geq 0 \text{ and } \sum_{i=1}^K P(C_i) = 1$$

choose C_i if $P(C_i | \mathbf{x}) = \max_k P(C_k | \mathbf{x})$

$$p(c_1|x) = \frac{p(x|c_1)p(c_1)}{p(x|c_1)p(c_1) + p(x|c_2)p(c_2)}$$



Losses and Risks

- Actions: α_i
- Loss of α_i when the state is C_k : λ_{ik}
- Expected risk (Duda and Hart, 1973)

$$R(\alpha_i | \mathbf{x}) = \sum_{k=1}^K \lambda_{ik} P(C_k | \mathbf{x})$$

choose α_i if $R(\alpha_i | \mathbf{x}) = \min_k R(\alpha_k | \mathbf{x})$

Losses and Risks: 0/1 Loss

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$$\lambda_{ik} = \begin{cases} 0 & \text{if } i = k \\ 1 & \text{if } i \neq k \end{cases}$$

$$\begin{aligned} R(\alpha_i | \mathbf{x}) &= \sum_{k=1}^K \lambda_{ik} P(C_k | \mathbf{x}) \\ &= \sum_{k \neq i} P(C_k | \mathbf{x}) \\ &= 1 - P(C_i | \mathbf{x}) \end{aligned}$$

For minimum risk, choose the most probable class

Losses and Risks: Reject

$$\lambda_{ik} = \begin{cases} 0 & \text{if } i = k \\ \lambda & \text{if } i = K + 1, \quad 0 < \lambda < 1 \\ 1 & \text{otherwise} \end{cases}$$

$$R(\alpha_{K+1} | \mathbf{x}) = \sum_{k=1}^K \lambda P(C_k | \mathbf{x}) = \lambda$$

$$R(\alpha_i | \mathbf{x}) = \sum_{k \neq i} P(C_k | \mathbf{x}) = 1 - P(C_i | \mathbf{x})$$

choose C_i if $P(C_i | \mathbf{x}) > P(C_k | \mathbf{x}) \quad \forall k \neq i$ and $P(C_i | \mathbf{x}) > 1 - \lambda$
reject otherwise

Given the loss table

Action	Truth	
	c_1	c_2
α_1 Choose c_1	0	10
α_2 Choose c_2	5	0

$$R(\alpha_1|x) = 0 \cdot P(c_1|x) + 10 \cdot P(c_2|x) \\ = 10(1 - P(c_1|x)) =$$

$$R(\alpha_2|x) = 5 \cdot P(c_1|x) + 0 \cdot P(c_2|x)$$

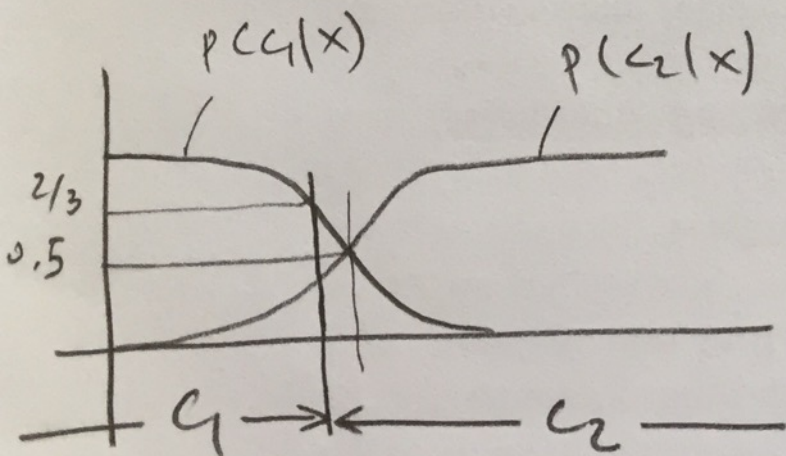
Choose α_1 (if)

$$R(\alpha_1|x) < R(\alpha_2|x)$$

$$10 - 10P(c_1|x) < 5P(c_1|x)$$

$$10 < 15 \cdot P(c_1|x)$$

$$P(c_1|x) > 10/15 = 2/3$$



Assigning c_2 to c_1 is costly, so choose c_1 only when very sure.

loss. table

Action	Truth	
	C_1	C_2
α_1 Choose C_1	0	10
α_2 Choose C_2	5	0
α_3 Reject	1	1

$$R(\alpha_1|x) = 10 \cdot P(C_2|x) = 10 - 10P(C_1|x)$$

$$R(\alpha_2|x) = 5 \cdot P(C_1|x)$$

$$R(\alpha_3|x) = 1$$

Choose α_1 if $P(C_1|x) > 2/3$

$$\text{and } 10 - 10P(C_1|x) < 1$$

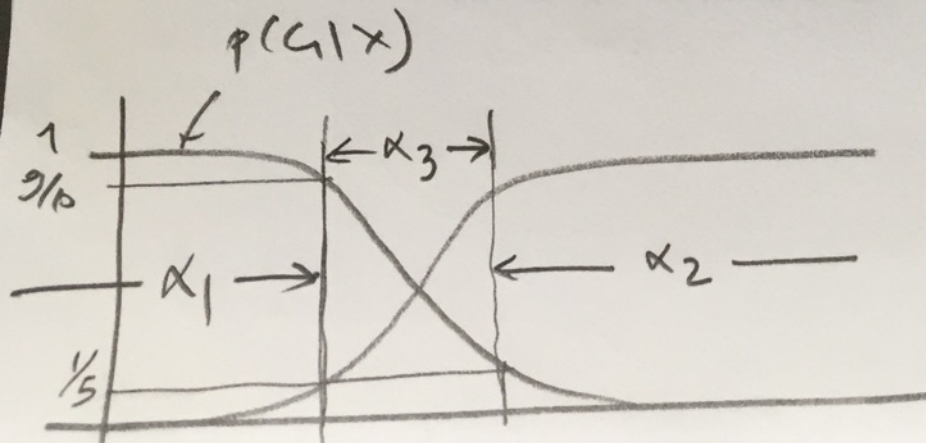
$$10P(C_1|x) > 9$$

$$P(C_1|x) > 9/10$$

Choose α_2 if $P(C_1|x) < 1/3$

$$\text{and } 5P(C_1|x) < 1$$

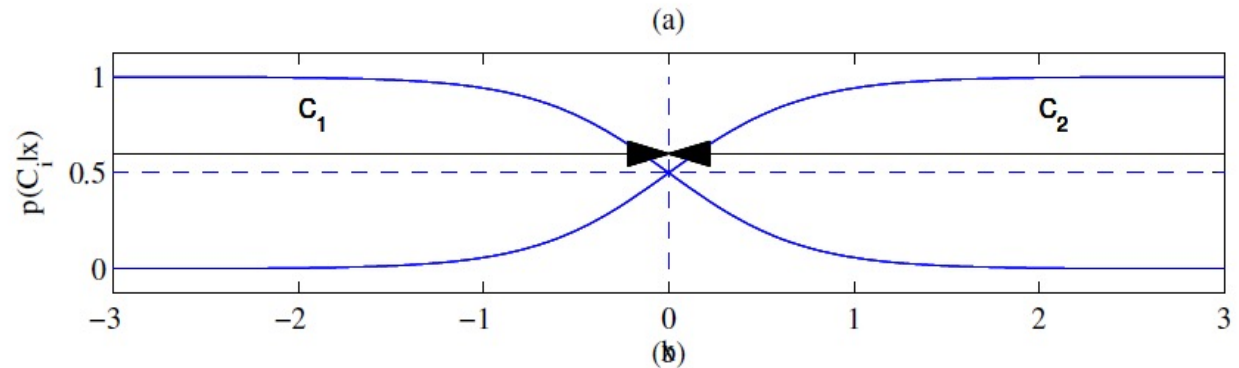
$$P(C_1|x) < 1/5$$



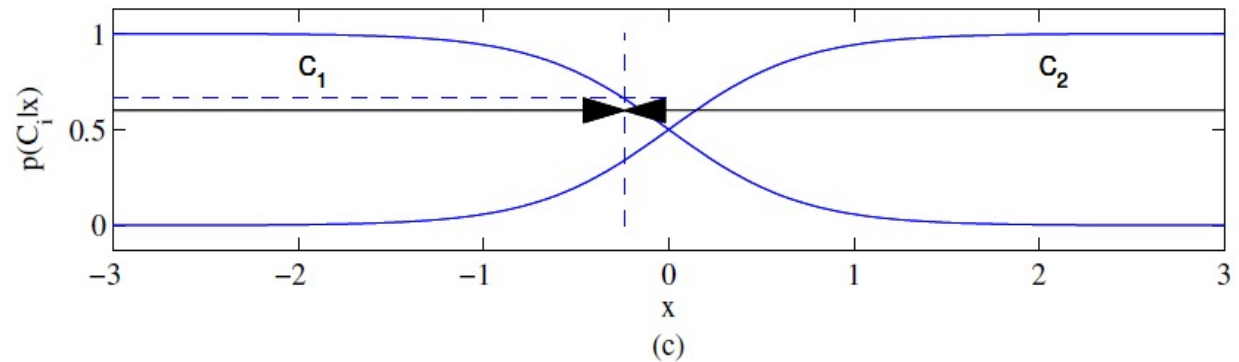
Different Losses and Reject

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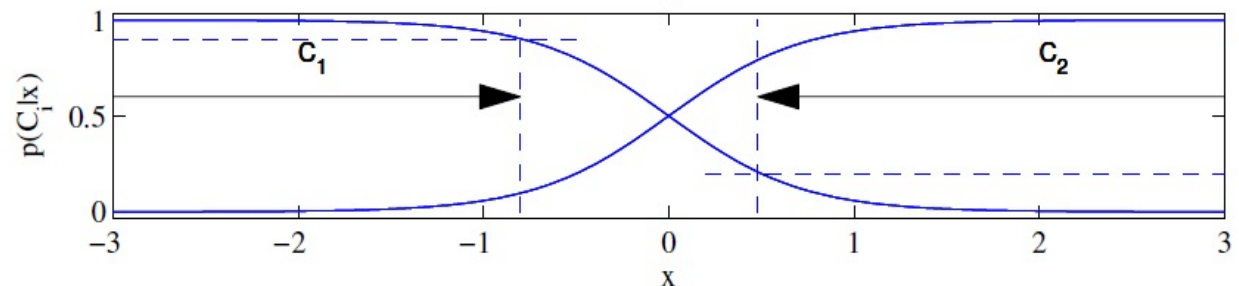
Equal losses



Unequal losses



With reject



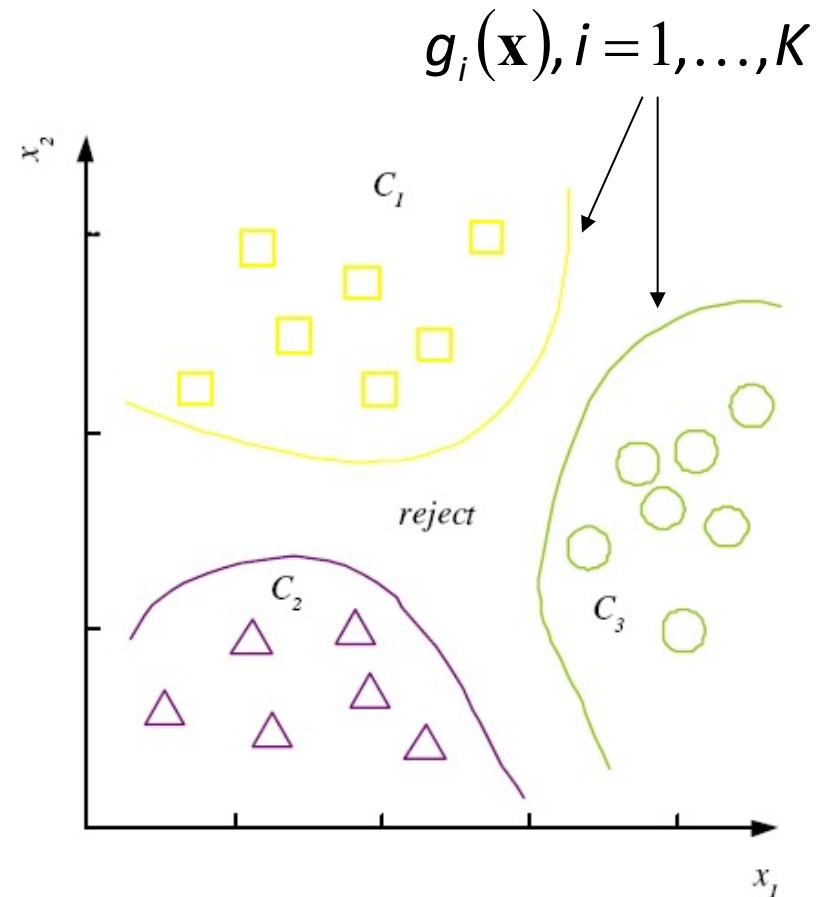
Discriminant Functions

choose C_i if $g_i(\mathbf{x}) = \max_k g_k(\mathbf{x})$

$$g_i(\mathbf{x}) = \begin{cases} -R(\alpha_i | \mathbf{x}) \\ P(C_i | \mathbf{x}) \\ p(\mathbf{x} | C_i)P(C_i) \end{cases}$$

K decision regions $\mathcal{R}_1, \dots, \mathcal{R}_K$

$$\mathcal{R}_i = \{\mathbf{x} | g_i(\mathbf{x}) = \max_k g_k(\mathbf{x})\}$$



K=2 Classes

□ Dichotomizer ($K=2$) vs Polychotomizer ($K>2$)

□ $g(\mathbf{x}) = g_1(\mathbf{x}) - g_2(\mathbf{x})$

choose $\begin{cases} C_1 & \text{if } g(\mathbf{x}) > 0 \\ C_2 & \text{otherwise} \end{cases}$

□ *Log odds:* $\log \frac{P(C_1 | \mathbf{x})}{P(C_2 | \mathbf{x})}$

Association Rules

- Association rule: $X \rightarrow Y$
- *People who buy/click/visit/enjoy X are also likely to buy/click/visit/enjoy Y.*
- A rule implies association, not necessarily causation.

Association measures

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- Support ($X \rightarrow Y$):

$$P(X, Y) = \frac{\#\{\text{customers who bought } X \text{ and } Y\}}{\#\{\text{customers}\}}$$

- Confidence ($X \rightarrow Y$):

$$P(Y | X) = \frac{P(X, Y)}{P(X)}$$

- Lift ($X \rightarrow Y$):
$$= \frac{P(X, Y)}{P(X)P(Y)} = \frac{P(Y | X)}{P(Y)}$$
$$= \frac{\#\{\text{customers who bought } X \text{ and } Y\}}{\#\{\text{customers who bought } X\}}$$

Example

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Transaction	Items in basket
1	milk, bananas, chocolate
2	milk, chocolate
3	milk, bananas
4	chocolate
5	chocolate
6	milk, chocolate

milk $\rightarrow$ bananas: Support: $\frac{\# \{ \text{milk, bananas} \}}{N} = \frac{2}{6} P(\text{milk} \wedge \text{bananas})$

Confidence: $\frac{\# \{ \text{milk, bananas} \}}{\# \{ \text{milk} \}} = \frac{2}{4} P(\text{bananas} | \text{milk})$

bananas $\rightarrow$ milk Confidence: $\frac{\# \{ \text{milk, bananas} \}}{\# \{ \text{bananas} \}} = \frac{2}{2} = 1.$

Apriori algorithm (Agrawal et al., 1996)

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- For (X,Y,Z) , a 3-item set, to be frequent (have enough support), (X,Y) , (X,Z) , and (Y,Z) should be frequent.
- If (X,Y) is not frequent, none of its supersets can be frequent.
- Once we find the frequent k -item sets, we convert them to rules: $X, Y \rightarrow Z, \dots$
and $X \rightarrow Y, Z, \dots$